

Markov Reward Processes: Computing State Values

J.C. Vaught

1 Introduction

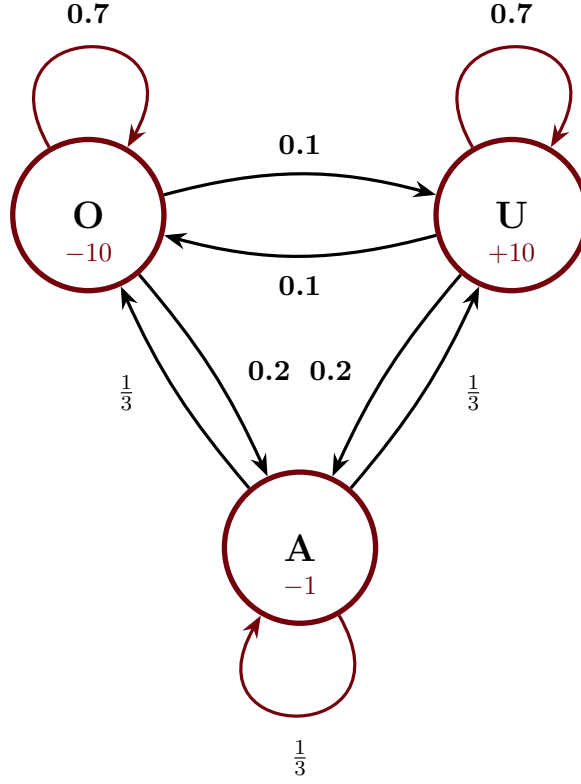
A **Markov Reward Process (MRP)** extends a standard Markov chain by associating a reward with each state. This framework is fundamental in reinforcement learning, economics, and operations research, where we want to evaluate the long-term value of being in a particular state.

Definition 1 (Markov Reward Process). A Markov Reward Process is a tuple $\langle \mathcal{S}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ where:

Symbol	Name	Description
$\mathcal{S}$	State space	Finite set of states
$\mathcal{P}$	Transition matrix	$\mathcal{P}_{ss'} = P(S_{t+1} = s' \mid S_t = s)$
$\mathcal{R}$	Reward function	$\mathcal{R}_s = \mathbb{E}[R_{t+1} \mid S_t = s]$
γ	Discount factor	$\gamma \in [0, 1]$

2 Problem Setup

Consider the following three-state Markov Reward Process:



2.1 State Space and Rewards

The state space is $\mathcal{S} = \{O, U, A\}$ with immediate rewards:

State	Name	Reward	Interpretation
O	Overweight	-10	Penalty
U	Underweight	+10	Reward
A	Average	-1	Small penalty

2.2 Transition Probability Matrix

The transition dynamics are captured by the matrix $\mathcal{P}$:

$$\mathcal{P} = \begin{pmatrix} P(O|O) & P(U|O) & P(A|O) \\ P(O|U) & P(U|U) & P(A|U) \\ P(O|A) & P(U|A) & P(A|A) \end{pmatrix} = \begin{pmatrix} 0.7 & 0.1 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

Note that each row sums to 1, as required for a valid probability distribution.

3 The Value Function

Definition 2 (State Value Function). The **value** of a state s is the expected cumulative discounted reward starting from state s :

$$V(s) = \mathbb{E} \left[\sum_{t=0}^{\infty} \gamma^t R_{t+1} \mid S_0 = s \right]$$

The discount factor γ determines how much we value future rewards:

Value	Meaning
$\gamma = 0$	Only immediate rewards matter
$\gamma = 1$	Future rewards equally valuable (may diverge)
$\gamma \in (0, 1)$	Geometrically decreasing weight on future

3.1 The Bellman Equation

The value function satisfies a recursive relationship known as the **Bellman equation**. In lecture, this was presented in its compact expectation form:

$$V(s) = R(s) + \gamma \mathbb{E}[V(S_{t+1}) \mid S_t = s]$$

While one could derive the value function using the formulation provided in class, I found it to be unwieldy and unfriendly to beginners. The expectation notation, though mathematically elegant, obscures the actual computation required. Therefore, I am utilizing the **expanded summation form**:

$$V(s) = R(s) + \gamma \sum_{s' \in \mathcal{S}} P(s'|s) \cdot V(s')$$

where:

Symbol	Meaning
$V(s)$	Value of state s (what we want to compute)
$R(s)$	Immediate reward received in state s
γ	Discount factor (how much we care about future)
s'	Each possible next state
$P(s' s)$	Probability of transitioning from s to s'
$V(s')$	Value of the next state s'

Remark 1 (Equivalence of Forms). These two forms are mathematically identical. The expectation $\mathbb{E}[V(S_{t+1}) \mid S_t = s]$ is simply the probability-weighted sum over all possible next states:

$$\mathbb{E}[V(S_{t+1}) \mid S_t = s] = \sum_{s' \in \mathcal{S}} P(s'|s) \cdot V(s')$$

The expanded form makes explicit *how* to actually compute the expectation, which is far more useful when solving problems by hand.

This equation states that the value of a state equals:

1. The **immediate reward** $R(s)$, plus
2. The **discounted expected value** of the next state (summing over all possibilities)

4 Hand Derivation

Let us solve for $V(O)$, $V(U)$, and $V(A)$ with discount factor $\gamma = 0.9$.

4.1 Step 1: Write the Bellman Equations

Applying the Bellman equation to each state:

$$\begin{aligned} V(O) &= R(O) + \gamma \left[P(O|O) \cdot V(O) + P(U|O) \cdot V(U) + P(A|O) \cdot V(A) \right] \\ &= -10 + 0.9 \left[0.7 \cdot V(O) + 0.1 \cdot V(U) + 0.2 \cdot V(A) \right] \end{aligned} \tag{1}$$

$$\begin{aligned} V(U) &= R(U) + \gamma \left[P(O|U) \cdot V(O) + P(U|U) \cdot V(U) + P(A|U) \cdot V(A) \right] \\ &= +10 + 0.9 \left[0.1 \cdot V(O) + 0.7 \cdot V(U) + 0.2 \cdot V(A) \right] \end{aligned} \tag{2}$$

$$\begin{aligned} V(A) &= R(A) + \gamma \left[P(O|A) \cdot V(O) + P(U|A) \cdot V(U) + P(A|A) \cdot V(A) \right] \\ &= -1 + 0.9 \left[\frac{1}{3} \cdot V(O) + \frac{1}{3} \cdot V(U) + \frac{1}{3} \cdot V(A) \right] \end{aligned} \tag{3}$$

Notice how the expanded form makes it immediately clear what to plug in—no mysterious expectations, just probabilities and values.

4.2 Step 2: Rearrange to Standard Form

Expanding and collecting terms:

From equation (1):

$$\begin{aligned}V(O) &= -10 + 0.63 \cdot V(O) + 0.09 \cdot V(U) + 0.18 \cdot V(A) \\V(O) - 0.63 \cdot V(O) &= -10 + 0.09 \cdot V(U) + 0.18 \cdot V(A) \\0.37 \cdot V(O) &= -10 + 0.09 \cdot V(U) + 0.18 \cdot V(A)\end{aligned}$$

From equation (2):

$$\begin{aligned}V(U) &= 10 + 0.09 \cdot V(O) + 0.63 \cdot V(U) + 0.18 \cdot V(A) \\0.37 \cdot V(U) &= 10 + 0.09 \cdot V(O) + 0.18 \cdot V(A)\end{aligned}$$

From equation (3):

$$\begin{aligned}V(A) &= -1 + 0.3 \cdot V(O) + 0.3 \cdot V(U) + 0.3 \cdot V(A) \\0.7 \cdot V(A) &= -1 + 0.3 \cdot V(O) + 0.3 \cdot V(U)\end{aligned}$$

4.3 Step 3: Matrix Formulation

The system can be written in matrix form as:

$$\mathbf{V} = \mathbf{R} + \gamma \mathcal{P} \mathbf{V}$$

Rearranging:

$$(\mathbf{I} - \gamma \mathcal{P}) \mathbf{V} = \mathbf{R}$$

Therefore:

$$\boxed{\mathbf{V} = (\mathbf{I} - \gamma \mathcal{P})^{-1} \mathbf{R}}$$

Let us compute $\mathbf{I} - \gamma \mathcal{P}$:

$$\begin{aligned}\mathbf{I} - 0.9\mathcal{P} &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - 0.9 \begin{pmatrix} 0.7 & 0.1 & 0.2 \\ 0.1 & 0.7 & 0.2 \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \\ &= \begin{pmatrix} 0.37 & -0.09 & -0.18 \\ -0.09 & 0.37 & -0.18 \\ -0.3 & -0.3 & 0.7 \end{pmatrix}\end{aligned}$$

4.4 Step 4: Solve the System

Computing the inverse and multiplying by $\mathbf{R} = (-10, 10, -1)^T$:

Using Cramer’s rule or matrix inversion, we obtain:

$ \begin{aligned} V(O) &\approx -23.75 \\ V(U) &\approx +32.01 \\ V(A) &\approx +1.04 \end{aligned} $

4.5 Step 5: Interpretation

- $V(O) \approx -23.75$: Starting in state O is bad. The immediate reward is -10 , and the system tends to stay in O (70% self-loop), accumulating more negative rewards.
- $V(U) \approx +32.01$: Starting in state U is highly valuable. The immediate reward is $+10$, and the system tends to stay in U, accumulating positive rewards.
- $V(A) \approx +1.04$: Starting in state A is slightly positive. Despite the -1 immediate reward, there’s a $\frac{1}{3}$ chance of transitioning to the valuable state U.

5 Solution Algorithms

5.1 Method 1: Direct Solution (Matrix Inversion)

For small state spaces, we can directly compute:

$$\mathbf{V} = (\mathbf{I} - \gamma\mathcal{P})^{-1}\mathbf{R}$$

Algorithm 1: Direct Solution via Matrix Inversion

Input: Transition matrix $\mathcal{P}$, reward vector $\mathbf{R}$, discount factor γ

Output: Value function $\mathbf{V}$

- 1: $\mathbf{A} \leftarrow \mathbf{I} - \gamma\mathcal{P}$ // Compute coefficient matrix
- 2: $\mathbf{V} \leftarrow \mathbf{A}^{-1}\mathbf{R}$ // Solve linear system
- 3: **return** $\mathbf{V}$

Complexity: $O(n^3)$ for n states (matrix inversion).

5.2 Method 2: Iterative Policy Evaluation

For large state spaces, iterative methods are more practical:

Algorithm 2: Iterative Policy Evaluation**Input:** Transition matrix $\mathcal{P}$, reward vector $\mathbf{R}$, discount factor γ , threshold θ **Output:** Value function $\mathbf{V}$

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1: Initialize  $V(s) \leftarrow 0$  for all  $s \in \mathcal{S}$ 
2: repeat
3:    $\Delta \leftarrow 0$ 
4:   for each state  $s \in \mathcal{S}$  do
5:      $v \leftarrow V(s)$  // Store old value
6:      $V(s) \leftarrow R(s) + \gamma \sum_{s'} P(s'|s) \cdot V(s')$  // Bellman update
7:      $\Delta \leftarrow \max(\Delta, |v - V(s)|)$  // Track max change
8:   end for
9: until  $\Delta < \theta$  // Convergence check
10: return  $\mathbf{V}$ 

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Complexity: $O(n^2)$ per iteration, typically converges in $O\left(\frac{1}{1-\gamma} \log \frac{1}{\theta}\right)$ iterations.

5.3 Convergence Guarantee

The iterative method is guaranteed to converge because the Bellman operator is a **contraction mapping** with factor $\gamma < 1$. By the Banach fixed-point theorem:

$$\|V_{k+1} - V^*\| \leq \gamma \|V_k - V^*\|$$

where V^* is the true value function.

6 Worked Example: Iterative Computation

Let us trace through three iterations of the iterative method with $\gamma = 0.9$:

Iteration 0: Initialize $V_0(O) = V_0(U) = V_0(A) = 0$

Iteration 1:

$$V_1(O) = -10 + 0.9[0.7(0) + 0.1(0) + 0.2(0)] = -10$$

$$V_1(U) = +10 + 0.9[0.1(0) + 0.7(0) + 0.2(0)] = +10$$

$$V_1(A) = -1 + 0.9[\frac{1}{3}(0) + \frac{1}{3}(0) + \frac{1}{3}(0)] = -1$$

Iteration 2:

$$V_2(O) = -10 + 0.9[0.7(-10) + 0.1(10) + 0.2(-1)] = -10 + 0.9(-6.2) = -15.58$$

$$V_2(U) = +10 + 0.9[0.1(-10) + 0.7(10) + 0.2(-1)] = +10 + 0.9(5.8) = +15.22$$

$$V_2(A) = -1 + 0.9[\frac{1}{3}(-10) + \frac{1}{3}(10) + \frac{1}{3}(-1)] = -1 + 0.9(-0.33) = -1.30$$

Iteration 3:

$$V_3(O) = -10 + 0.9[0.7(-15.58) + 0.1(15.22) + 0.2(-1.30)] \approx -18.83$$

$$V_3(U) = +10 + 0.9[0.1(-15.58) + 0.7(15.22) + 0.2(-1.30)] \approx +18.35$$

$$V_3(A) = -1 + 0.9[\frac{1}{3}(-15.58) + \frac{1}{3}(15.22) + \frac{1}{3}(-1.30)] \approx -1.50$$

The values continue to converge toward the analytical solution as iterations progress.

Iteration	V(O)	V(U)	V(A)
0	0.00	0.00	0.00
1	-10.00	+10.00	-1.00
2	-15.58	+15.22	-1.30
3	-18.83	+18.35	-1.50
$\vdots$	$\vdots$	$\vdots$	$\vdots$
∞	-23.75	+32.01	+1.04

Table 1: Convergence of iterative policy evaluation ($\gamma = 0.9$)

7 Summary

1. A **Markov Reward Process** associates rewards with states in a Markov chain.
2. The **value function** $V(s)$ measures expected cumulative discounted reward from state s .
3. The **Bellman equation** provides a recursive relationship. While the lecture presented it as:

$$V(s) = R(s) + \gamma \mathbb{E}[V(S_{t+1})|S_t = s]$$

the expanded form is far more practical for computation:

$$V(s) = R(s) + \gamma \sum_{s'} P(s'|s) V(s')$$

4. Solutions can be computed via:

- **Direct method:** Matrix inversion (exact, $O(n^3)$)
- **Iterative method:** Repeated Bellman updates (approximate, scalable)

5. For our three-state MRP with $\gamma = 0.9$:

$$V(O) \approx -23.75, \quad V(U) \approx +32.01, \quad V(A) \approx +1.04$$